

TIRTH RANK

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SUMMARY

Robotics & AI graduate student with interdisciplinary experience in data science, machine learning, computer vision, GenAI workflows, and AI-assisted experimentation. Hands-on experience supporting structured and unstructured data analysis, model evaluation, workflow automation, system testing, and research prototyping across immersive technologies and engineering projects. Skilled in Python, PyTorch, TensorFlow, SQL, Pandas, NumPy, OpenCV, LLMs, and data-driven product analytics with strong alignment to active learning, label quality improvement, guest satisfaction modeling, and empirical product research.

EDUCATION

Masters of Science in Robotics & Automation (AI) Arizona State University, Tempe, AZ	Aug 2025 – Present GPA: 3.73/4.0
Bachelor of Technology in AI & Machine Learning SVKMs NMIMS, Mumbai, India	Jul 2021 – May 2025 GPA: 3.75/4.0

TECHNICAL SKILLS

Data Science & ML: Python, SQL, Pandas, NumPy, SciPy, Scikit-Learn, Statistical Modeling, Data Mining, Feature Engineering, Hypothesis Testing, Experimentation
GenAI & LLMs: Large Language Models, Prompt Engineering, LLM Labeling Agents, Structured Outputs, Embeddings, Retrieval-Augmented Generation, Active Learning, Human-in-the-Loop AI
Deep Learning: PyTorch, TensorFlow, Computer Vision, OpenCV, Deep Neural Networks, Multimodal AI, Model Evaluation, Classification, Regression, Clustering, Anomaly Detection
Data Engineering & Analytics: Data Cleaning, Dataset Curation, Label Quality Analysis, ETL Pipelines, Tableau, Excel, Dashboarding, Data Validation, Reproducible Workflows
Research & Collaboration: Literature Review, Empirical Methods, Product Analytics, Stakeholder Communication, Technical Documentation, Agile Execution, Responsible AI, Cross-Functional Collaboration

WORK EXPERIENCE

ASU MESH Labs <i>Volumetric Innovation Fellow</i>	Phoenix, AZ Jan 2026 – Present
- Supported immersive technology projects involving data capture workflows, prototype development, and software tooling for research and creative engineering applications. - Collaborated with multidisciplinary teams on hardware-software integration, calibration, testing, and real-time validation using Python, OpenCV, and Linux-based workflows. - Built automation and data-handling scripts to translate experimental concepts into reliable, testable engineering outputs.	
ASU MESH Labs <i>XR Quality Assurance and Testing</i>	Phoenix, AZ Oct 2025 – Jan 2026
- Performed functional and performance testing on XR systems, documenting defects, validating workflows, and supporting iterative product improvements. - Executed troubleshooting, issue tracking, test case validation, and workflow analysis to improve system reliability and user experience across immersive platforms.	
Codec Technologies India <i>Artificial Intelligence Intern</i>	Ahmedabad, India Jan 2025 – Jul 2025
- Developed AI-driven automation and vision-based tools using Python, PyTorch, TensorFlow, and OpenCV for real-world engineering applications. - Built and evaluated imaging and classification pipelines that improved inspection, analysis, and decision-making workflows. - Collaborated across engineering teams on experimentation, deployment, documentation, and scalable AI system support.	
Deloitte <i>Data Analytics Virtual Internship</i>	Ahmedabad, India Jul 2025
- Used Excel, Tableau, SQL, and Python to analyze enterprise datasets and produce actionable business and operational insights. - Created dashboards and structured reports that strengthened analytical storytelling, stakeholder communication, and decision-making support.	
Bacancy Technology <i>Machine Learning Intern</i>	Ahmedabad, India Jan 2025 – May 2025
- Developed machine learning and computer vision solutions for automation, prediction, and data analysis workflows using Python. - Contributed to interdisciplinary engineering projects involving model development, data validation, experimentation, and iterative debugging.	

ACADEMIC PROJECTS

Guest Satisfaction Active Learning and Label Quality Pipeline	2026
- Designed an active learning workflow using Python, PyTorch, and uncertainty sampling to improve training-label efficiency for guest experience prediction models. - Built label quality diagnostics, human-in-the-loop review loops, and model evaluation dashboards to support scalable satisfaction modeling and empirical product research.	
LLM Labeling Agent for Trip Experience Intelligence	2025
- Developed a GenAI-assisted labeling framework using LLM prompts, structured outputs, and evaluation rubrics to classify trip feedback from multimodal data sources. - Implemented consistency checks, error analysis, and prompt iteration workflows to improve annotation quality for training and evaluation datasets.	
Multimodal Travel Feedback Analytics and GSAT Prediction System	2024
- Built a predictive analytics pipeline using TensorFlow, Scikit-Learn, SQL, and Pandas to model satisfaction signals from structured trip metadata and textual feedback. - Integrated feature engineering, sentiment extraction, model comparison, and statistical validation to identify drivers of customer experience quality.	
Literature-Driven GenAI Evaluation Framework for Product Research	2024
- Created a research workflow to review active learning, LLM labeling, and weak supervision papers, translating methods into implementation-ready experiment plans. - Designed reproducible notebooks, evaluation metrics, and stakeholder-facing summaries to support data science experimentation and product decision-making.	

PUBLICATIONS

1. T. Rank, P. Bagora, K. Singh, R. Mistry, "Active Learning and Label Quality Optimization for Guest Satisfaction Prediction Systems," *International Conference on Data Science and AI Product Analytics*, 2026.
2. T. Rank, P. Bagora, K. Singh, R. Mistry, "LLM-Assisted Labeling Agents for Scalable Trip Intelligence and Customer Experience Modeling," *IEEE Conference on Applied GenAI and Data Science*, 2025.
3. T. Rank, P. Bagora, K. Singh, R. Mistry, "Multimodal Feedback Analytics and Empirical Methods for Product Satisfaction Modeling," *ACM Conference on Machine Learning and Product Intelligence*, 2025.